Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

Bab 1

PENDAHULUAN

Bab 2

TINJAUAN PUSTAKA

styles/fbe$_t ez[utf8x]inputencamsmath, amsthm, amssymb[bottom]footmisccitegraphicxlongtable$
multirow subfigure algorithm algorithmic

THESIS TITLE TEZ BAŞLIĞI B.S., Program Name, Boğaziçi University, 2010 Name Surname Your Program 2016

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Acknowledgements come here...

Ringkasan

One page abstract will come here.

Bir sayfa uzunluğunda özet gelecektir.

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a_{ij}Description of a_{ij} $\mathbf{A}$State transition matrix of a hidden Markov model αBlending parameter *or* scale $\beta_t(i)$Backward variable ΘParameter set

2DTwo Dimensional 3DThree Dimensional AAMActive Appearance Model ASMActive Shape Model

Bab 3

INTRODUCTION

Start with an introduction...

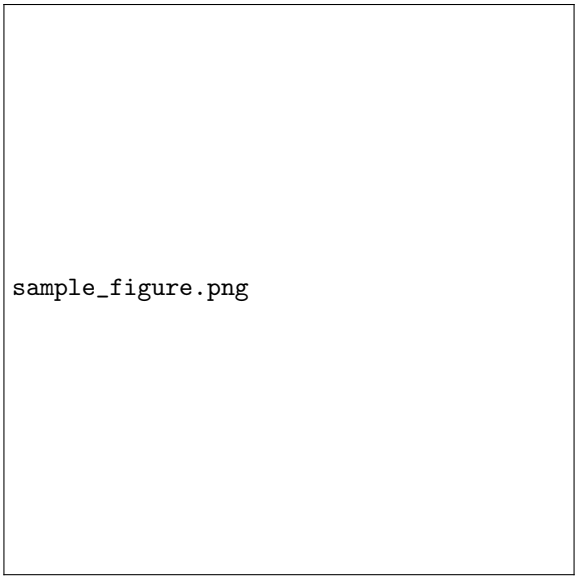
Bab 4

EXPERIMENTS AND RESULTS

Experiments and results come here...

4.1 Sample Section

Always place some text after headings before putting a graphics into a section as seen in Figure 4.1.



Gambar 4.1: Sin and Cosine.

Now, let us cite some studies: one source as [**?**], two sources as [**?**, **?**] or you may cite three or more sources as [**?**, **?**, **?**]. Observe that they are ordered in the references chapter in the same

order as they are cited. Let us put a sample table as seen in Table 4.1. Please pay attention that the caption is followed by a period.

Tabel 4.1: Sample table.

	Header 1	**Header 2**
Row 1	Bla bla bla	Bla bla bla
Row 2	Bla bla bla	Bla bla bla

Footnotes should be avoided as possible. If there is an absolute necessity, footnotes should be used as this.[1]

Item lists may be represented as follows:

- This is an item. Do not use boldface for the items.
    1. This is a sub-item. Subsub-items are not allowed.
- Another item.

Item lists may also be represented as follows:

1. This is another enumerated item.
    - This is another sub-item.

Theorem 4.1 *The solutions of the equation* $ax^2 + bx + c = 0$ *with* $a \neq 0$ *are*

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We use the method of completing the square to rewrite $ax^2 + bx + c$. $\mathrm{ax}^2 + bx + c = a\left(x^2 + \frac{b}{a}x+\right) + c$
$= a\left(x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 +\right) + c$
$= a\left(x + \frac{b}{2a}\right)^2 - a\left(\frac{b}{2a}\right)^2 + c$
$= a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2-4ac}{4a}$. Therefore $ax^2 + bx + c = 0$ can be rewritten as

$$a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a} = 0,$$

which can in turn be rearranged as

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

[1] Example of a footnote

Taking square roots gives

$$x + \frac{b}{2a} = \frac{\pm\sqrt{b^2 - 4ac}}{2a}$$

which implies

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

as required. Finally, we will put a sample algorithm (PCA algorithm) using the relevant package in a figure as shown in Figure 4.1 and sample equations.

Require $\mathbf{s_i}$, $i = 1, 2, \ldots, N$ are normalized
Compute the mean $\bar{\mathbf{s}}$ using Eq. 4.1; Form the $N \times 2L$ matrix $\mathbf{Q}$ as defined in Eq. 4.1; $N < 2 \times L$ $\mathbf{Q} \Leftarrow \mathbf{Q}^T$; Compute the covariance matrix $\mathbf{C}_s$ using Eq. 4.1;Decompose $\mathbf{C}_s$ to its eigenvectors $\mathbf{e}_k$ and eigenvalues λ_k satisfying Eq. 4.1; $N < 2 \times L$ $k = 1$ to K $\mathbf{e}_k \Leftarrow \mathbf{Q}\mathbf{e}_k$; $\mathbf{e}_k \Leftarrow \mathbf{e}_k / ||\mathbf{e}_k||$ (normalization);

Gambar 4.2: Principal Component Analysis Algorithm.

$\bar{s} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{s}_i$
$\mathbf{Q} = \left[\begin{array}{cccc} \mathbf{s}_1 - \bar{\mathbf{s}} & \mathbf{s}_2 - \bar{\mathbf{s}} & \cdots & \mathbf{s}_N - \bar{\mathbf{s}} \end{array} \right]_{2L \times N}$
$\mathbf{C}_s = \frac{1}{N} \mathbf{Q}^T \mathbf{Q}$

$$\mathbf{C}_s \mathbf{e}_k = \lambda_k \mathbf{e}_k \tag{4.1}$$

4.1.1 Example of First Subheadings

Some text here

Example of Second Subheadings

Some text here too.

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CONCLUSION

The conclusions of the thesis should come here.

Lampiran A

APPLICATION

The appendices start here.